QUANTA

THE CAPACITY MODEL™

QUANTA HUMAN PERFORMANCE PRINCIPLES
FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

TRAINEE MANUAL

V 1.0

NLC

THANK YOU

This course is the product of long months of research, discussion, planning, real-world application, more discussion, and site visits with workers and leaders in the utility industry worldwide. The key principles of The Capacity Model® are based on the research of international experts in workplace safety, human performance, risk management, and other fields. We are grateful for their outstanding contributions to the industry and for their cooperation with our team.

Our goal is to improve how all workers and executives approach safety in the workplace—and to bring more people home safely to their families every night.

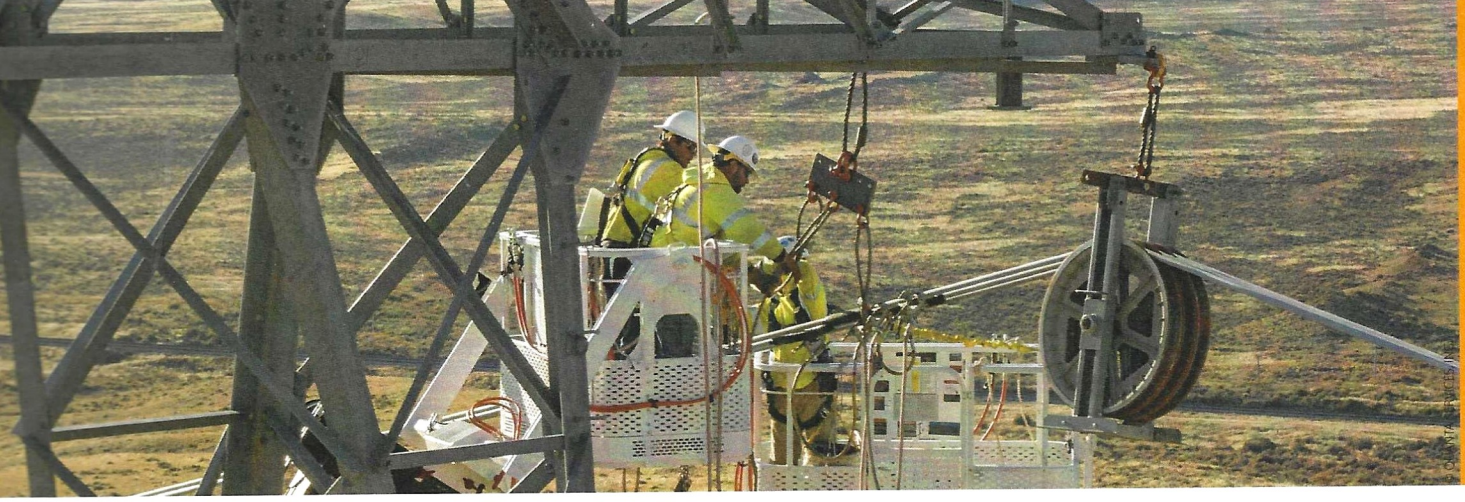
We'd like to thank the dedicated individuals and organizations who have devoted so much of their time and effort toward making this training a reality.

Quanta Services Corporate SH&E Department

Quanta Operating Companies

Individual Experts in Industrial Safety

THE CAPACITY MODEL™



CONTENTS

Syllabus 4

Agenda 5

Ch. 1 Rethinking Safety 7

1.1 A Paradigm Shift 8

1.2 A Three-Pillar Approach 11

1.3 Quanta Human Performance Principles 16

Activity: A Case Study 20

Activity: Accountability Scenarios 24

Activity: Productive Response 30

Glossary 35

Sources 37

Appendix 39

Quanta Human Performance Principles for the Power and Telecommunications Industry

© 2023 Northwest Lineman College.

All rights reserved. This book may not be duplicated in any way without the express written consent of Northwest Lineman College. The information contained herein is for the training or reference use of the reader and may not be incorporated in any commercial programs, other books, databases, web-related applications, or any kind of software without written consent of Northwest Lineman College. Making copies of this book, or any portion of this book, for any purpose is a violation of United States copyright law. Violators will be prosecuted.

Limits of Liability and Disclaimer of Warranty
Northwest Lineman College has used its best efforts in preparing this material and the content contained in it. These efforts include the development, research, and, in some cases, testing of the subjects to determine their effectiveness. Northwest Lineman College makes no warranty of any kind, expressed or implied, with regard to these subjects or the documentation contained in this book.

Northwest Lineman College shall not be liable in the event of incidental or accidental consequential damages in connection with, or arising out of, the furnishing, performance, or the use of the training, associated instructions, and/or claims of injury.

Trademarks
Trademarked names may appear throughout this book. Rather than list the names and entities that own the trademarks or insert a trademark symbol with each mention of the trademarked name, Northwest Lineman College states that it is using the names for editorial purposes only and to the benefit of the trademark owner, with no intention of infringing upon that trademark.

Northwest Lineman College

7600 S Meridian Rd, Meridian, ID 83642

P. 208-888-4817

F. 208-888-4275

Global Position: 43° 32' 9.70" N 116° 23' 34.43" W

www.lineman.edu

© 2023 Northwest Lineman College. All rights reserved.
TCM_HPP_PowerTel_Manual 20231026 2:29 PM

TCM 3001

HUMAN PERFORMANCE PRINCIPLES FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

SYLLABUS

COURSE DESCRIPTION

Trainees learn how the Quanta Human Performance Principles support The Capacity Model®, how to apply the learning approach to accountability, and act on events with a productive response. Trainees who complete the course taught by an NLC instructor will receive a certificate of completion from NLC.

The class is taken in conjunction with the STKY and the Energy Wheel module. Complete both modules within three months to gain maximum learning and retention benefits.

Intended audience: Operations management, safety professionals, supervisors who direct the work of others, and those responsible for leading job briefings.

LEARNING OUTCOMES

1.a. Explain what The Capacity Model® is.
1.b. Explain how the Quanta Human Performance Principles support The Capacity Model®.
1.c. Compare the learning and blameapproaches to accountability.
1.d. Select a productive response for a given outcome.

SUGGESTED PREREQUISITES

- Understanding The Capacity Model®

COURSE MATERIALS

- Materials to be issued at the beginning of the course.

INSTRUCTIONAL METHODS

- Lecture
- Learning activities
- Knowledge check

COURSE OUTLINE

- Housekeeping
- Quanta Human Performance Principles
- Exam

METHODS OF MEASUREMENT

KNOWLEDGE

Written exam:
A = 100%–90%
B = 89%–80%
C = 79%–70%
D = 69%–60%
F = 59% and below

If a student earns less than a 70% on the exam, one retake is permitted. If they do not earn at least a 70% on the exam retake, they must retake the course.

SKILL

Completion of all activities as directed by the instructor.

BEHAVIOR

Active participation and engagement in class discussions and activities.

STUDENT-TO-INSTRUCTOR RATIO
Maximum of 24:1.

CLOCK HOURS
3 Hours

POLICIES AND EXPECTATIONS

Building a positive learning environment—Be sure to share, support your classmates, and contribute to the positive learning environment during your course. You bring incredible value to the group and to the course. Please refrain from any behavior that you know doesn't work in a classroom environment.

TCM 3001

HUMAN PERFORMANCE PRINCIPLES FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

AGENDA

SCHEDULE

HOUSEKEEPING	*15 minutes*
QUANTA HUMAN PERFORMANCE PRINCIPLES	*2 hours 15 minutes*
EXAM	*30 minutes*

Breaks are scheduled at the discretion of the instructor.

The lunch schedule is determined by the class start time and the option to complete the STKY and the Energy Wheel module in the same day.

RETHINKING SAFETY

After completing this chapter, you should be able to:

1.a. Explain what The Capacity Model® is.

1.b. Explain how the Quanta Human Performance Principles support The Capacity Model®.

1.c. Compare the *learning* and *blame* approaches to accountability.

1.d. Select a productive response for a given outcome.

1.1 A PARADIGM SHIFT

Safety and health standards implemented during the last several decades have prevented countless work-related injuries and illnesses.[1] The number of life-ending workplace events has also declined but at a slower rate.[1] To further reduce catastrophic workplace events, Quanta Services has worked with industry-leading experts to develop The Capacity Model®. It is a new approach to the way we plan, execute, and learn from work that strives to eliminate life-threatening, life-altering, and life-ending events. This model shifts the focus from simply preventing mistakes to building the space or capacity for these mistakes to happen safely. By introducing the *capacity to fail safely* into the workplace, mistakes can be absorbed without major injury or death.

Employees must still work to prevent accidents. However, we can expand our approach to identify energies that, when released and not controlled, could be catastrophic. As employees put controls in place to protect against those energies, we can better fulfill our overall objective: bring everyone home safely.

Fig. 1.1. The Capacity Model can be applied to any industry.

THE ACCIDENT TRIANGLE

You may be familiar with Heinrich's Accident Triangle. For a long time, it has been a cornerstone of workplace health and safety philosophy.[2] Heinrich's theory suggested a relationship between different levels of industrial incidents. He proposed that reducing the number of unsafe acts would create a corresponding fall in the number of near misses, injuries, and fatalities. The theory has faced criticism for the values allocated to each category and for focusing mainly on reducing unsafe acts.

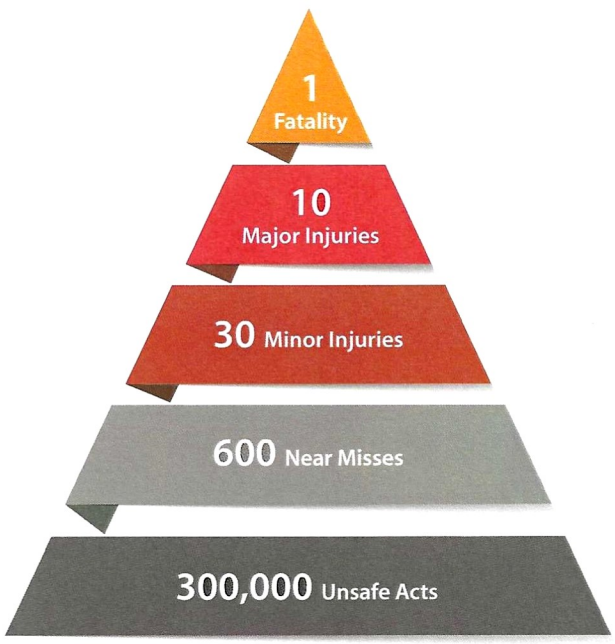


Fig. 1.2. Heinrich's Accident Triangle.

The problem with this model is that, while statistically accurate, it does not predict future events. In fact, data has shown that there is little to no correlation between acts that result in minor injuries and acts that result in major injuries or fatalities.[2]

For example, if a crew experienced 30 sprained ankles during a year, it would be fair to say they are not working safely. However, those injured ankles do not indicate that a fatality is sure to happen.[3] Similarly, if procedures were changed to eliminate the sprained ankles, doing so would not protect workers from a different type of life-threatening, life-altering, or life-ending injury—such as falling from an elevated surface.

Between 1972 and 2021, data from the U.S. Bureau of Labor Statistics showed that total reportable incident rates (TRIR) of injuries and illnesses dropped nearly 75% from 10.9 per 100 full-time equivalent workers to 2.7.[4,5] Fatality rates dropped 81% from 17.0 per 100,000 full-time equivalent workers in 1972 to about 3.3 in 2013.[6,7] However, data shows the rate of workplace fatalities are on the rise, increasing to 3.6 in 2021.[7]

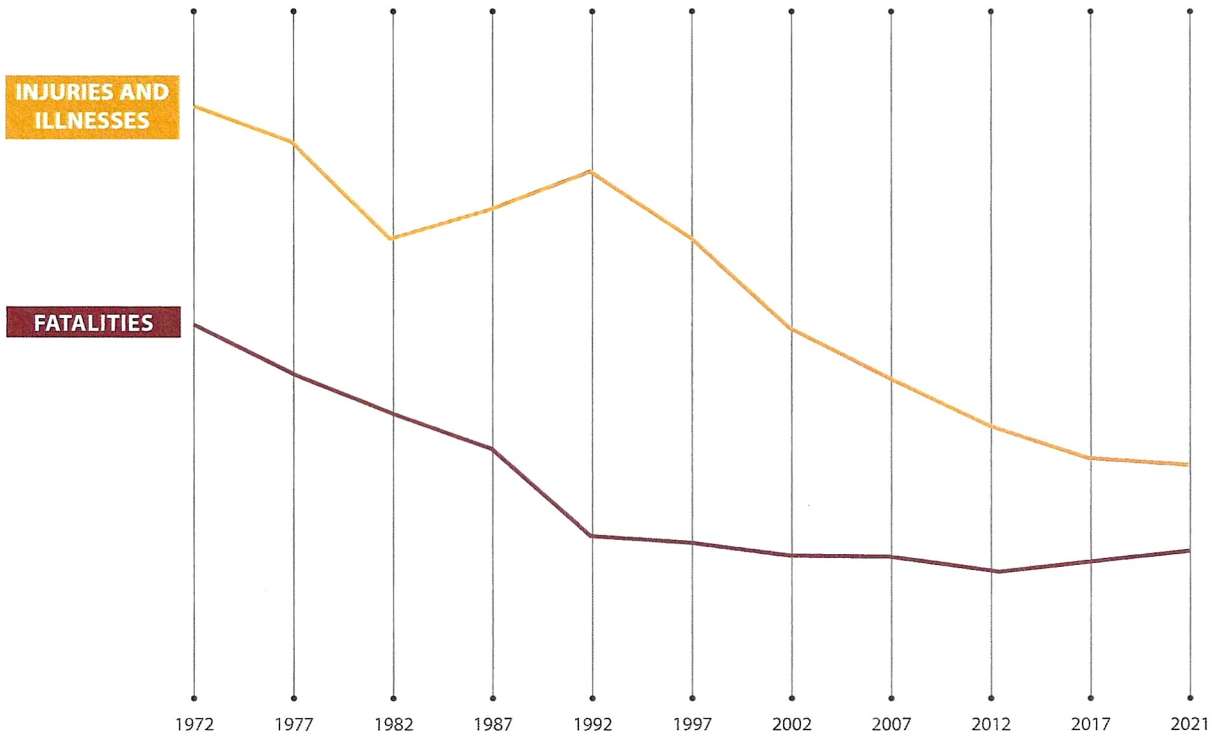


Fig. 1.3. Since the creation of OSHA, workplace injuries and illnesses have dropped significantly. After years of steady decline, the U.S. Bureau of Labor Statistics reports an increase in workplace fatalities. While injuries and illnesses far outnumber fatalities, this graph overlays both measurements to show the trend each is following.

1.2 A THREE-PILLAR APPROACH

The Capacity Model seeks to change the game by taking a three-pillar approach to safety: prevention, learning, and capacity for failure.

- **Prevention**: Continue to work at preventing unwanted events from occurring.
- **Learning**: Accept that mistakes are inevitable in life and business. Learn from them to improve prevention and build capacity.
- **Capacity for Failure**: Put controls in place that allow failure to occur safely and mitigate the consequences of an unwanted event.

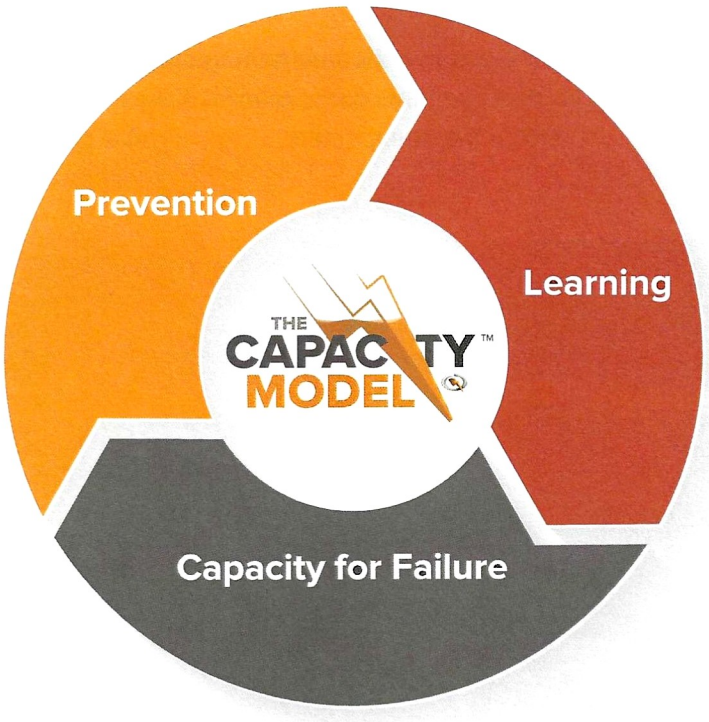


Fig. 1.4. This three-pillar approach to workplace safety applies to every work situation.

PREVENTION

Organizations and employees must continue to practice accident prevention. Figure 1.3 shows the dramatic improvement in workplace safety as a result of prevention. This is accomplished through **preventive controls**—those acts, objects, or systems designed to prevent an unwanted event. This can include following OSHA regulations and company standards, getting all necessary training, using equipment correctly, and employing other tactics that improve day-to-day safety.

Accident prevention is necessary, but it is not enough. Life-threatening, life-altering, or life-ending injuries occur when energies are released without proper **mitigative controls**—those acts, objects, or systems designed to reduce the consequences of an unwanted event. In Figure 1.5, preventive controls are put in place to prevent falls. Mitigative controls, like fall arrest systems, are also implemented to protect workers when they fall.

The Capacity Model is not about replacing prevention with mitigation but about shifting to a balance of the two. Continue to work at prevention but plan for failure as if it were going to happen, and then add controls to handle the unexpected.

Fig. 1.5. Incorporate preventive controls to prevent unwanted events. Add mitigative controls to protect against those events when they happen.

© QUANTA SERVICES, INC.

LEARNING

Traditional safety tends to look "down and in," meaning it seeks to understand *why* an event occurred rather than *how* it occurred. Who made the mistake? Why didn't they follow procedure? How can we "fix them" so it doesn't happen again? Accident investigations usually lead to some level of blame, training around the event, or a combination of both.

What's missing with the "down and in" view of safety is the valuable context that surrounds nearly every mistake. What led to this event? What warning signs did we fail to notice before it happened? Would a different worker have made a better decision? We need to focus on how processes or systems fail, not why people fail.

A surprising revelation about mistakes is that they usually happen during normal working conditions and are often made by experienced personnel.[1,7] No one plans to come to work and have an accident. Therefore, we must look more deeply at the factors that lead up to these incidents so we can understand *how* they are occurring. With that understanding, we can build the capacity needed to avoid injuries when incidents occur. This is known as the "up and out" approach of viewing safety.

The Traditional Safety Model (The Blame Approach)	**The New View of Safety** (The Learning Approach)
• Focuses on *why* workers failed. • Seeks to find blame and punish. • Asks: "*Why* did the event occur?" "*Who* made the mistake?" "How can we fix *them*, so it doesn't happen again?"	• Focuses on *how* instead of why a process or system failed. • Seeks to understand and learn. • Asks: "How did the event occur?" "What led to this event?" "What warning signs did we fail to notice before it happened?"

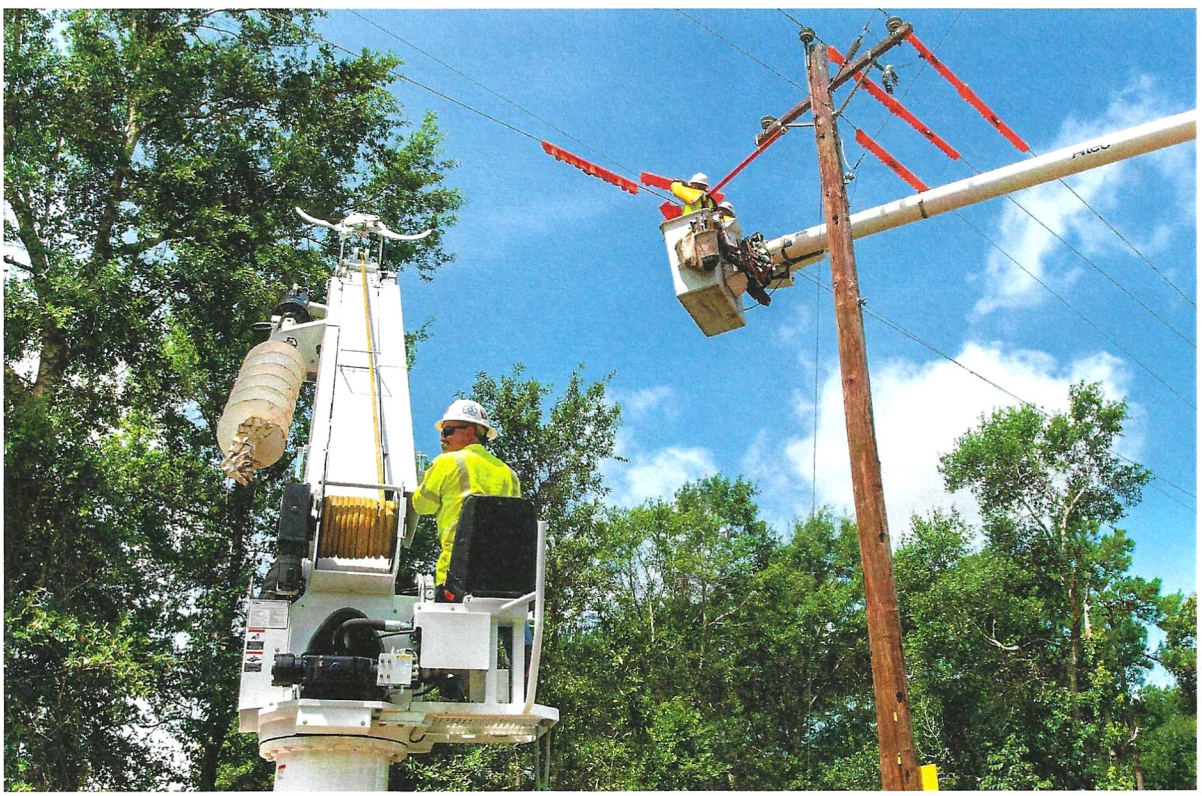
Fig. 1.6. The Capacity Model seeks to learn from mistakes to improve prevention and build capacity.

One benefit of the learning approach is that a company can apply the same tactics when things go well. Instead of only studying system failures after something bad happens, companies can use operational learning to examine a successful process. The data gathered through this positive research can uncover successful procedures that would help the entire organization. It also introduces employees to operational learning in a positive context.

CAPACITY FOR FAILURE

For this training, *capacity* is defined as "the actual or potential ability to perform, yield, or withstand." With The Capacity Model, **capacity** is the ability to perform work safely, yield safe results, and withstand the outcome of an accident or mistake. Having the capacity for failure at work means the operation is flexible enough to adapt to unexpected disruptions, absorb those disruptions safely, and then continue without experiencing a life-threatening, life-altering, or life-ending event.

Workers must assume an accident or mistake will happen. Even though they try to prevent it, something will go wrong. The question is, *how are we protected when that happens?* That's what building capacity for failure is all about.



© QUANTA SERVICES, INC.

Fig. 1.7. Building capacity for failure involves multiple layers of protection.

APPLYING THE CAPACITY MODEL

The Capacity Model features four distinct elements for creating capacity.

1. **STKY (Stuff That Kills You)**
   A focus on the hazards that can seriously injure or kill people.
2. **The Quanta Energy Wheel**
   A hazard-identification tool that should be used before and during work to identify hazards.
3. **Operational Learning**
   An approach to learning and improving that examines both successful operations and failure. It relies on frontline workers to explain how work is performed. It seeks to understand the context of an event in order to determine where system strengths and gaps exist, promote successful operations, and reduce the likelihood of negative events from repeating.
4. **STKY Controls Framework**
   The controls framework builds capacity for failure by providing a consistent, disciplined approach to identifying controls. The framework identifies the critical tasks that organizations perform and the minimum controls that must be in place before work begins.

Before learning about these elements, it is important to understand Quanta's seven human principles. They are the foundation for The Capacity Model.

© QUANTA SERVICES, INC.

Fig. 1.8. Applying the four elements of The Capacity Model helps crews safely navigate a complex and dynamic work environment.

1.3 QUANTA HUMAN PERFORMANCE PRINCIPLES

"Perfection is not attainable, but if we chase perfection we can catch excellence." -Vince Lombardi, NFL Hall of Fame coach

To drive down the rate of catastrophic events, we must do something different. Working to prevent unwanted events from happening has made a huge difference, but it is not enough. As an industry leader, Quanta Services is moving from a language of accidents and enforcement to a language of awareness and improvement: Safety is the presence of capacity, not the absence of unwanted events.

In the context of worker safety, **human performance** refers to the actual ways that people interact with their real-life work environment—their tasks, equipment, coworkers, physical setting, and culture. As workers deal with constantly changing duties, environments, and hazards, failure will occur. Those who embrace a human performance mindset find realistic methods for allowing this failure to occur safely because they understand how normal human behavior plays out in the workplace.

The Quanta human performance principles are not just a new set of rules or goals. There is no endpoint. Instead, they are levels of understanding that will take time and deliberate action to embed in a company's culture. Human performance is fundamental to The Capacity Model and to improving safety.

Fig. 1.9. The Quanta human performance principles.

1 PEOPLE MAKE MISTAKES.

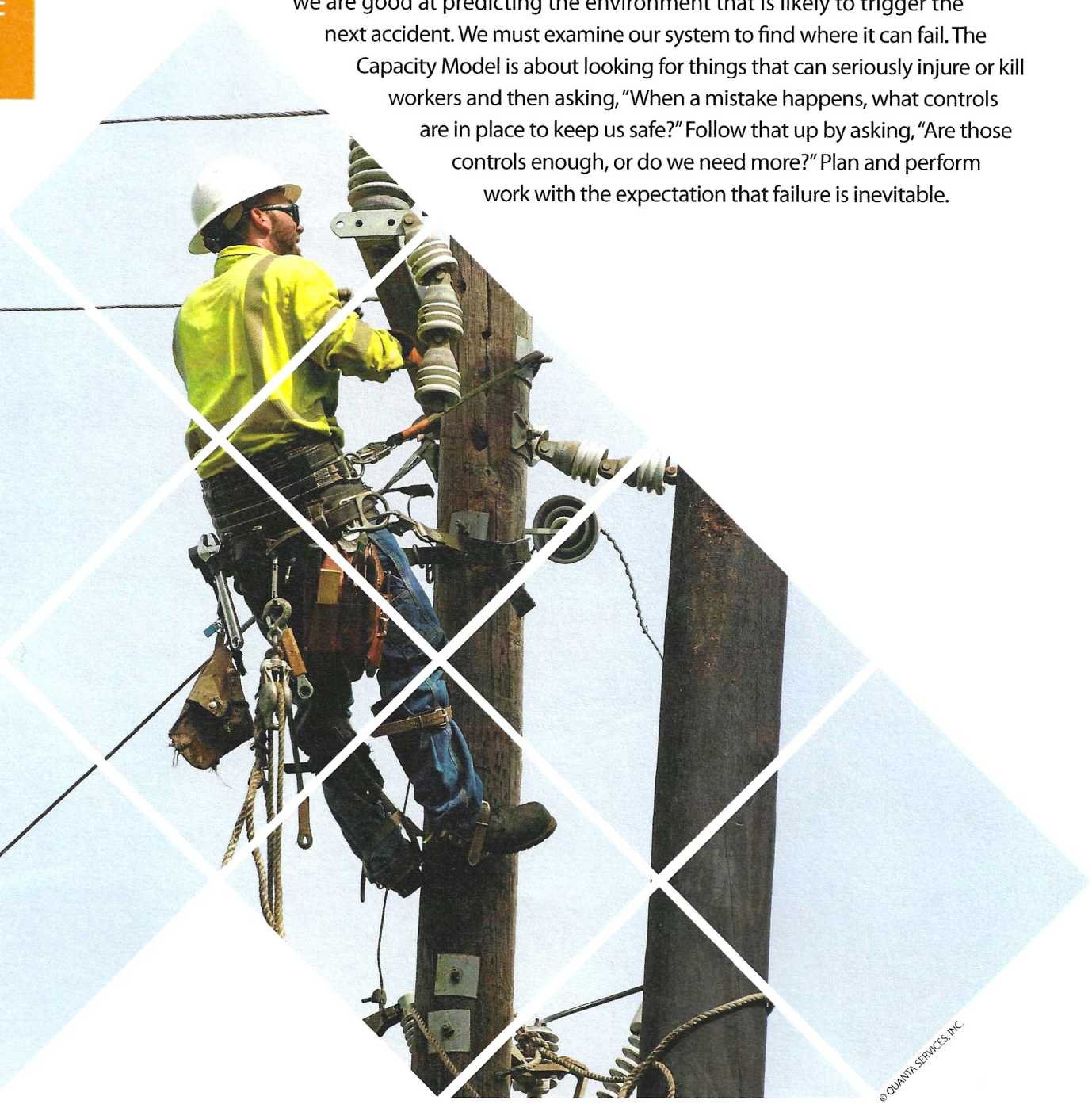
Error is normal. Even the best people make mistakes. After an unexpected event, it's easy to think, "Workers just need to work more safely." But if we approach a situation with the understanding that error is normal, we will build systems that can absorb these errors. Machines are great for dull, repetitive processes. However, more complex work requires the emotion, experience, and creativity only humans can provide. The trade-off is that the worker may not get it right the first time (or the second, third, or fourth time). For all sorts of reasons, failure will eventually happen. Let this knowledge be a rich source of learning and potential growth.

Fig. 1.10. This crew has built capacity into their work to allow failures to occur safely.

2 FAILURES CAN OCCUR SAFELY.

It's not *if*, it's *when*. We will always plan and execute our work as if failure is going to happen today.

People are not good at predicting where the next accident will happen, but we are good at predicting the environment that is likely to trigger the next accident. We must examine our system to find where it can fail. The Capacity Model is about looking for things that can seriously injure or kill workers and then asking, "When a mistake happens, what controls are in place to keep us safe?" Follow that up by asking, "Are those controls enough, or do we need more?" Plan and perform work with the expectation that failure is inevitable.



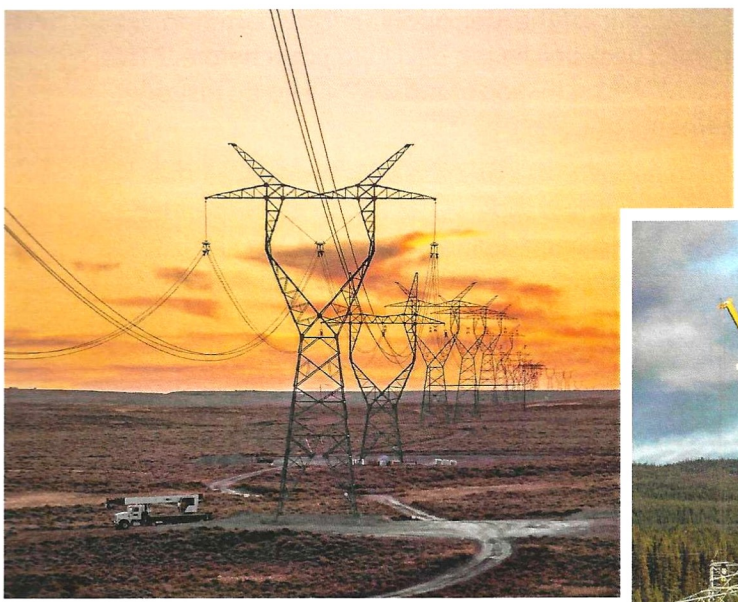
© QUANTA SERVICES, INC.

Fig. 1.11. What controls are in place to keep this worker safe?

3 CONTEXT INFLUENCES BEHAVIOR.

Workers do what they do for a reason, and the reason makes sense to the worker, given the context of the situation. The old belief is that if a worker who made a mistake is removed from the system, the system improves. Unfortunately, doing so also removes a huge opportunity to learn why a mistake happened. Understanding the context around how work is done can only be achieved by involving those who actually do the work.

A worker's real-time perspective of an event is very different from an after-the-event investigation and report focused on creating a linear description of events. We generally see well-meaning behaviors intended to get the job done despite a dynamic work environment. To learn the context of an unwanted event—and how those conditions impacted the worker—we must shift our thinking from "Who failed?" to "What failed?" and, more importantly, "How did our system let that happen?" Context is everything.



© QUANTA SERVICES, INC.

Fig. 1.12. No two jobs are alike. While the tasks may be the same, the crews must navigate varying and dynamic environments as they execute work plans.

"HUMAN ERROR" OR FLAWED CULTURE?

READ THE CASE STUDY AND RESPOND TO THE QUESTIONS.

Shortly before 10:00 AM on Tuesday, June 27, 2017, two subway cars of New York City's A train left the tracks as they approached the 125th Street station.[8]

The violent stop sent passengers flying. At least 34 people were injured from the derailment or from smoke inhalation caused when garbage in the tunnel was ignited during the crash.[9]

This was bad news for Joseph Lhota, chairman of the Metropolitan Transportation Authority (MTA), who was in his first week back on the job he'd held years before.[10] This rail accident would eventually cost the MTA $3.4 million in repairs[11]—and be just one of seven NY-area rail crashes or derailments during 2017.[12]

The preliminary investigation moved swiftly. Within a day, the MTA announced that "human error" caused the derailment and that two supervisors had been suspended. The night before the accident, work crews had been replacing damaged tracks. Instead of carting out the leftover track segments when they left, the crews stowed them on the track bed to be picked up later by a work train. These loose rails triggered the passing train's emergency brakes, causing the derailment.[11, 13]

Is storing leftover equipment on the track recklessly irresponsible? "Storing equipment between tracks is a common practice employed by railroads across the country to accelerate rail repairs," Lhota said in a later statement. "The key to this being an effective and safe practice is making sure that the extra equipment is properly bolted down, which does not appear to have happened in this case."[9] Further, the track segment was 6 ft shorter than the rules permitted.[13]

But John Samuelsen, president of the union that represents subway workers (though not the suspended supervisors), points out that it was management's responsibility to remove the extra rails. He noted that storing equipment this way had been done for decades and simply calling it "human error" was "oversimplifying the tragedy." Michael Carrube, who represents the supervisors, decried the swift placing of blame: "It's horrendous to point the finger and take someone out of service prior to the completion of the full investigation."[13] One commentator questioned how effective it is to suspend or replace workers if the system stays the same.[14]

1. Have you experienced or seen "swift investigations" that ended with "human error" as the cause? Give an example.

2. What did the company learn by suspending the two supervisors?

3. How effective is it to suspend or replace workers if the system stays the same?

PHOTO BY FIRST ENERGY CORP.

Fig. 1.13. When unwanted events occur, leaders must take a learning approach to understand what failed and allow the workers to be a part of the solution .

4 VIOLATIONS ARE RARELY, IF EVER, MALICIOUS.

Blame fixes nothing. Anger and blame may feel productive in the moment, but they do little to improve safety. Worse, these reactions discourage people from elevating near misses or mistakes in the future. "You can blame and punish or learn and improve, but you can't do both."[15]

Employees generally do their work correctly and safely. Willful, malicious violations intended to hurt or destroy are extremely rare. However, workers are often expected to follow operational rules that do not fit the reality of the job. So they adjust their behavior to fit the circumstances. If a worker knew their action would have a terrible consequence, they would have made a different choice. When we think about an unwanted event, we cannot assume a better worker would have made a better choice.

But wait. Employees still need to be held accountable, right?

That's correct—if "holding them accountable" means something other than just punishment. When an unwanted event happens, people often decide that *accountability* suddenly means only *fault that must be punished.* That is a shallow, unproductive definition.

Accountability is an obligation to accept responsibility and account for or explain one's actions. In a work setting, there are two approaches to accountability: learning and blame. *Learning* is forward-looking and focused on understanding the context surrounding an event. A worker is obligated to give their account of what happened and be part of the solution to help prevent the incident from repeating. On the other hand, *blame* is backward-looking and short-sighted, seeking to find fault and punish. Knowing that error is normal but malicious violations are rare, workers must be held accountable but not automatically punished every time something goes wrong.

ACTIVITY

ACCOUNTABILITY SCENARIOS

READ EACH SCENARIO AND CIRCLE THE ACCOUNTABILITY APPROACH THAT IS REPRESENTED IN THE SCENARIO. REMEMBER, *LEARNING* LOOKS FOR WAYS TO IMPROVE SYSTEMS AND PROCESSES, WHILE *BLAME* LOOKS TO FIND FAULT AND PUNISH.

1. An employee working 25 ft. up did not attach his lanyard to the anchor system. He lost his footing trying to catch a piece of hardware that had slipped from his grip. He lost his balance and fell. When he landed, he had a compound fracture that severed an artery, and required the use of a tourniquet to control bleeding. He was rushed to the hospital where he received blood transfusions to replace the lost blood. The safety bulletin cited "failed to follow procedure," "failed to recognize the hazard," and "human error" as the root of the cause.

   LEARNING BLAME

2. An employee working in an aerial lift lost his balance while performing a pole transfer. He leaned against the basket door. The basket door latch failed to catch, and the worker forgot to attach the safety chain. He fell from the basket but was caught by his fall arrest system. The employee sustained bruises from the harness and general soreness, but otherwise needed no further medical attention. The foreman generated a near miss report and sent it to his manager. The manager created a safety flash and alerted all crews in her district emphasizing how the door latch, safety chain, and fall restraint all built in capacity to fail safely.

   LEARNING BLAME

3. An employee was the safety watch for a reel of strand being back-pulled through a rear easement. The crew had to make a sudden stop while pulling out the strand. She was wearing appropriate PPE but broke her hand, trying to stop the reel from overspin. Her management team asked her to complete the zero-accident pledge again. They also asked her to present a complacency refresher during the Monday morning safety talk.

   LEARNING BLAME

4. Three employees were opening a trench looking for a break in an underground cable. Using a backhoe, the operator opened a 2 ½ ft X 9 ft trench and found the cable. One of the workers jumped into the unbarricaded trench to quickly dig around the location where the trouble was measured. A local foreperson driving by directed the crew to stop work and had the worker exit the trench. Soon after the worker left the trench, it collapsed. The foreperson initiated a full-crew discussion about what happened. He asked questions such as, "How did we get to a place where we executed work without shoring?" and "After we install shoring, what else do we need?"

   LEARNING BLAME

5. A forklift struck an employee because the operator was distracted by a radio call. As a result, the employee suffered a crushed foot that led to a permanent limp that medical treatment could not correct. All employees participated in a mandatory "safety stand-down." The operator received a written warning and was issued a formal disciplinary letter in their personnel file.

   LEARNING BLAME

6. The rigging of a suspended load (a 1,300-lb pole) broke, causing the load to crash to the ground just a few feet away from a crewmember. There was a marked exclusion zone, but the worker had not seen it. He was uninjured. The local management team pulled their crews together for a one-hour discussion in which they asked:
   - "Are the procedures for lifting and rigging practical and doable? Help me understand."
   - "How do you normally mark exclusion zones?"
   - "Where can we get better? What do you suggest?"

   LEARNING BLAME

7. A telecom worker was pulling cable when the pulling line broke, whipping back and hitting her and a colleague. She was not wearing any PPE and was standing in the line of fire. Management terminated the worker on the spot for not following the procedure, and the operations manager sent the crew home for two days without pay.

   LEARNING BLAME

8. An employee was working in an aerial lift to tension strand. While the strand was being put under tension, one of the spans became caught under an untrimmed branch. When the pressure on the strand released from the branch, it traveled upward and contacted power lines—causing the strand to become momentarily energized. The employee working the lift was not wearing properly rated rubber gloves while tensioning the strand and received an electrical shock resulting in third degree burns on his left hand and forearm. The manager joined the crew at the local hospital with the safety training documents for strand placement in hand, fired the injured employee on the spot, and showed the rest of the crew their signatures on the safety training documents.

   LEARNING BLAME

9. A telecom worker slipped while working on a tower and fell, but his fall arrest system caught him. He had a few scrapes and bruises but was able to climb back down safely. His management team led a discussion with the crew and asked, "What are some areas that are tricky to navigate? Any ideas on how to make them easier?"

   LEARNING BLAME

10. An employee was working on a 10-ft ladder, which another worker stabilized from below. The worker on the ladder accidentally dropped a tool that hit the lower worker in the face. Injuries included a fractured orbital bone and permanent vision loss in her right eye. The management team conducted a discussion with the workers and asked, "How do we normally protect ourselves in this type of situation? Is there an area that we can improve? Help me understand how to reduce this risk in the future."

    LEARNING BLAME

5 INCIDENTS CAN STEM FROM NORMAL DEVIATIONS.

How we imagine work takes place is different from how work actually takes place. Incidents do not happen because everything fails. They happen because safeguards are not in place, not working, or not effective at protecting against an unexpected combination of events. Knowing that people make mistakes, crews must design their work to have capacity to fail safely.

All work will deviate from the plan. Especially in the complex utility industry, workers must constantly adapt to changing circumstances. We call this the black-line/blue-line effect (see fig. 1.14). Work as planned (black line) never actually matches work as it is done (blue line). Expect the work being done to be uncertain and surprising. Workers must account for this inconsistency to maintain enough margin for error—capacity—to fail safely.

Hazard identification does not end after the first job briefing of the day. Throughout the work day, crews must be hyper-focused on identifying hazards (the red line) anytime there is a change in conditions or scope (the blue line). By identifying hazards and the unwanted events that can stem from them, crews can make sure that preventive and mitigative controls are in place and functioning when events occur.

If we want to learn from incidents, we need to move away from looking at why the work deviated from the plan (black line vs. blue line). Instead, we need to focus on how the work is performed. If we understand the hazards associated with the work, we can identify the controls that must be in place.

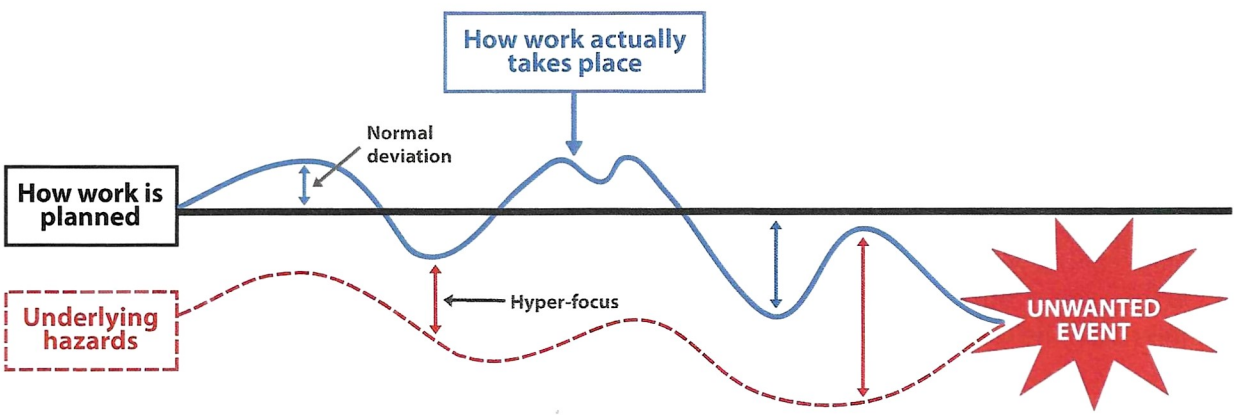


Fig. 1.14. How work is planned (black line) rarely matches how work is actually done (blue line).

6 MANAGEMENT'S RESPONSE TO FAILURE MATTERS.

Managers shape how the organization learns by their response to failure. A leader's response signals to employees whether it is safe to talk about workplace realities—hazards, mistakes, near misses, and even actual events. Management's responses are observed, felt, mimicked, and will spread until they define a company's culture. The same can be said for management's response to success.

Discipline should not be the immediate response when something bad happens. Remember the difference between the *learning* (forward-looking improvement) and *blame* (backward-looking punishment) approaches to accountability. By leaping to penalties too quickly, employers lose the ability to learn the context that allowed the incident to occur. Every failure presents an opportunity to learn. Whether or not a leader takes advantage of that opportunity lies in how they respond to that failure.

© QUANTA SERVICES, INC.

Fig. 1.15. Managers shape a company's culture by how they react to daily situations.

© QUANTA SERVICES, INC.

Fig. 1.16. Workers want leaders who demonstrate a desire to learn how the crew navigates the reality of the job site.

The way a manager should respond varies as much as the type of incident. But there are some basic steps to follow, remembering that a crew will feed off the behavior of their leader.

1. *Breathe and think.* During an unwanted event, it's normal to focus on yourself: your wrecked schedule, the blemish on your safety record, and the trouble you may be in. This natural, knee-jerk, "autopilot" reaction is usually followed by anger that someone else's mistake affects you. Shut off that autopilot, pause the anger, and think clearly about what is actually happening. Do not speak or even think the word "fault" because you have higher priorities right now.
2. *Resolve any emergencies.* Shut off the machinery, clear the danger area, look for the injured, get medical help rolling (whether that's retrieving a first aid kit or calling 911), and strengthen your defenses (protect your crew from hazardous domino effects).
3. *Adapt to the new reality.* Don't pretend that nothing has happened. The job site is a different place than it was only minutes before, and your crew is in a different place mentally. Even if no one was hurt and your equipment is undamaged, be realistic about what needs to happen next and how best to get it done.
4. *Follow your processes.* Do you need to call corporate? Do you need to complete a report? Do you need to inspect some equipment? Do you need to hold a follow-up job briefing? Is there a handbook you should be following? These processes were created to guide distracted workers through challenging, stressful situations.

YOUR RESPONSE MATTERS

The words a leader uses following an event greatly influence how workers respond to the same event. While a supervisor's panic, fear, and rage will spread to the crew, so will steady precision, calm urgency, and clear thinking. It boils down to the difference between reacting and responding to an incident.

A reaction is an immediate, unconscious behavior driven by an emotional trigger, such as anger or surprise. Reactions tend to be rash, unfiltered, and unproductive. On the other hand, a response is an intentional action based on a conscious decision. It requires self-control and enough time to steady your emotions and think critically about "what is the productive thing to do right now?"[16]

- *Unproductive reaction:* Quick, emotional, tense, and provokes emotional reactions from others. Thoughtless reactions move away from learning.
- *Productive response:* Takes time, requires thought, brings calm, and invites a discussion. Thoughtful responses move toward learning.

EVENT	UNPRODUCTIVE REACTIONS	PRODUCTIVE RESPONSES
A worker was just hit by falling equipment.	• Who's not paying attention? • Is it a recordable? • There goes my schedule. • Is the equipment broken?	• Is the worker OK? • What is the current condition of the site—is everyone safe and secure?
A trench just collapsed.	• Then start digging again. • Why can't you guys do this right for once? • Now when will the project get done?	• Was there anyone in the trench? • Is the site stable or do we need to clear the scene? • What steps need to happen before we start work again?
A worker had a near miss with a backing vehicle.	• @#$%* idiot! • Were they following procedure? • Why'd they do that?	• Tell me the story of how this happened. • How can we improve our traffic control procedures to keep this from repeating? • Help me understand what led to this.
An underground utility line was accidentally cut.	• Just fix it. • I bet I know what happened. • Which rule did you break?	• Who do we need to call? • What factors led up to this event? • What else do I need to know about what happened?

Table 1.1. The differences between unproductive reactions and productive responses.

ACTIVITY

PRODUCTIVE RESPONSE

FOR EACH SCENARIO, CHOOSE THE PRODUCTIVE RESPONSE. REMEMBER, A PRODUCTIVE RESPONSE MOVES TOWARD LEARNING, WHILE AN UNPRODUCTIVE REACTION MOVES AWAY FROM LEARNING.

© QUANTA SERVICES, INC.

Select the best immediate response—the first thing you should say.

SCENARIO ONE

An employee climbed into the bed of the digger derrick to retrieve a shovel. She didn't use three points of contact and fell from the top step.

a. Is she okay?
b. Is it a recordable?
c. Why did this happen?
d. What needs to happen to this employee to make her follow proper procedure?

SCENARIO TWO

An employee was working on a 3-ft high ledge when he fell, resulting in a bruised hip and a cut that required three stitches.

a. Everybody back to work.
b. I bet I know what happened, and who it was.
c. Is the facility safe, secure, and stable?
d. What do we need to do so that he pays better attention?

SCENARIO THREE

A forklift operator was loading empty reels into the back of a truck. The truck driver, not realizing his vehicle was still being loaded, tried to pull away. The loading dock restraint system stopped the truck from moving. The forklift operator was able to drive safely out of the truck bed.

a. Were they following procedures?
b. Is this an opportunity to learn?
c. Why didn't the truck driver check first?
d. Did anything break? What will be the cost?

SCENARIO FOUR

A crew was working near the edge of an unbarricaded, unshored trench. As a truck drove past, the ground beneath one of the workers caved in and buried him in the trench. Luckily the worker was okay.

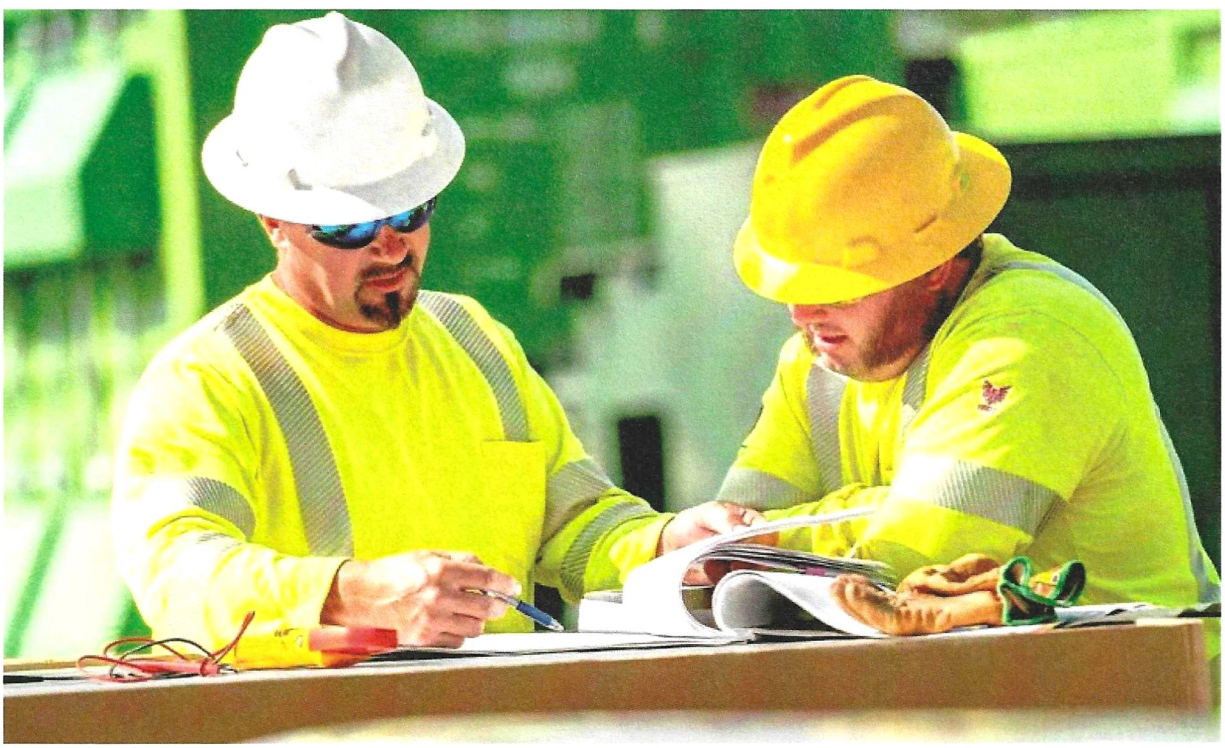
a. What is the root cause?
b. What corrective action do we need to take with the crew?
c. When can we start this work again?
d. Where else on the job site could this happen?

7 LEARNING IS A DELIBERATE IMPROVEMENT STRATEGY.

Learning is a strategic and operational choice toward improvement. A quick, reactionary change on the heels of an unexpected outcome may feel like a fix but won't necessarily make the company safer. Such "easy" reactions often overpower a company's desire to learn context and make a careful decision. Taking the time to learn how work is actually done demonstrates to workers a commitment to true safety improvement.

Company leaders who make an effort to learn—and repeatedly demonstrate *productive responses* to events, good and bad—will change the organization in a positive way. Why pay attention to the good as well as the bad? Because when "nothing is happening," everything is happening. Remember that error is normal, so if everything is perfect, we need to either learn the source of that success or uncover the problems hidden from view.

Are your workers good or just lucky? And how do you know?



© QUANTA SERVICES, INC.

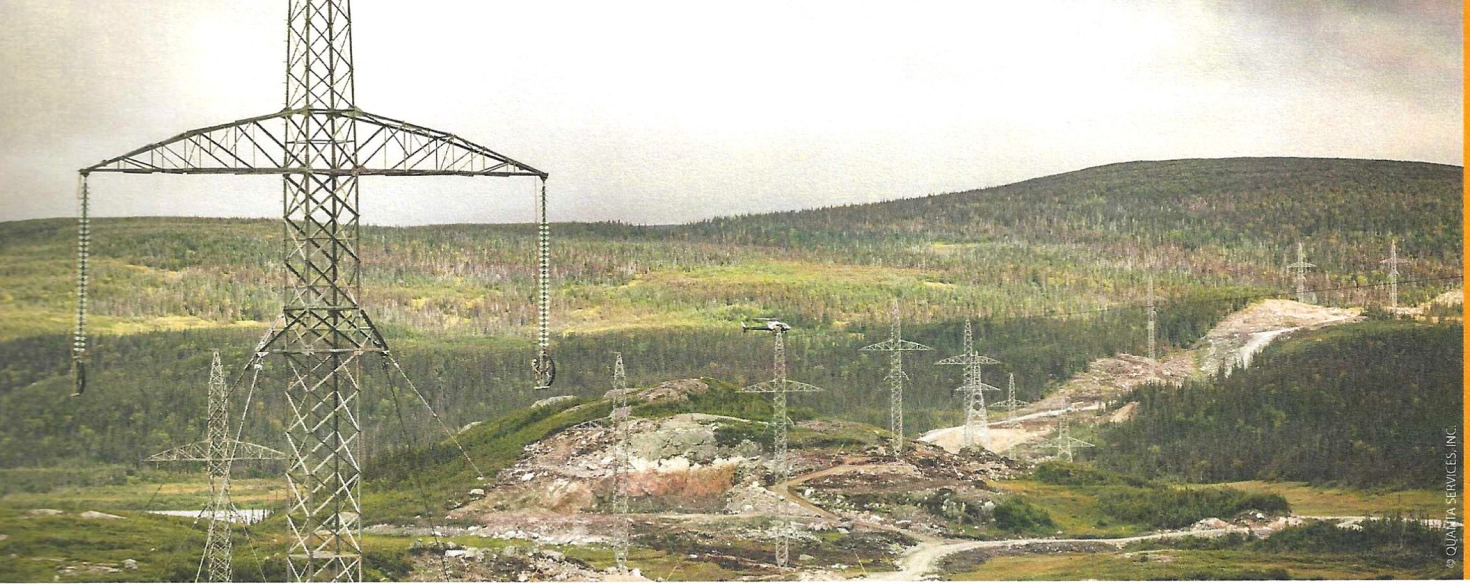
Fig. 1.17. Companies can learn just as much from a "perfect" job as from an accident.

NOTES

NOTES

GLOSSARY

accountability	An obligation to accept responsibility and account for or explain one's actions.
capacity	The actual or potential ability to perform, yield, or withstand. With The Capacity Model, the ability to perform work safely, yield safe results, and withstand the outcome of an accident or mistake.
capacity for failure	In a work setting, the ability to perform a task while safely withstanding the outcome of an accident or mistake.
human performance	In the context of worker safety, the actual ways that people interact with their real-life work environment—their tasks, equipment, coworkers, physical setting, and culture. It accounts for how normal human behavior plays out in the workplace.
operational learning	An approach to learning about the context of an event to promote successful operations and prevent negative events from repeating. For example, instead of firing someone who made a mistake, find out the circumstances that caused that person to make the mistake—and then fix those circumstances.
mitigative controls	Acts, objects, or systems designed to reduce the consequences of an unwanted event. Mitigative controls help after the bad stuff has already happened.
preventive controls	Acts, objects, or systems designed to prevent an unwanted event. Preventive controls stop the bad stuff from happening.
reaction	An immediate, unconscious behavior driven by an emotional trigger, such as anger or surprise.
response	An intentional action based on a conscious decision.



SOURCES

CONTENT

1. OSHA.gov (2020). All About OSHA. Retrieved from https://www.osha.gov/sites/default/files/publications/all_about_OSHA.pdf.

2. Johnson, Ashley (2011, Oct. 1). Examining the foundation: Were Heinrich's theories valid, and do they still matter? *Safety + Health*. Retrieved from https://www.safetyandhealthmagazine.com/articles/6368-examining-the-foundation.

3. Harris, Ned (guest), and Todd Conklin (host). (2019, November 16, #248). *PreAccident Investigation Podcast*. Retrieved from https://www.podbean.com/ew/pb-m4z72-c6587f.

4. Brown, Jeff. U.S. Bureau of Labor Statistics (2020, July). Beyond the Numbers: Nearly 50 years of occupational safety and health data. Retrieved from https://www.bls.gov/opub/btn/volume-9/nearly-50-years-of-occupational-safety-and-health-data.htm.

5. U.S. Bureau of Labor Statistics. (2022, Nov. 09). Employer-Reported Workplace Injuries and Illnesses, 2021. Retrieved from: https://www.bls.gov/charts/injuries-and-illnesses/total-nonfatal-work-injuries-and-illnesses-by-year.htm# and https://www.bls.gov/news.release/osh.nr0.htm

6. AFL-CIO Safety and Health Office. (2021, May). Death on the Job The Toll of Neglect: A National and State-by-State Profile of Worker Safety and Health in the United States. Retrieved from https://aflcio.org/sites/default/files/2021-05/DOTJ2021_Final.pdf.

7. U.S. Bureau of Labor Statistics (2022, Dec. 16). National Census of Fatal Occupational Injuries in 2021.Retrieved from https://www.bls.gov/news.release/pdf/cfoi.pdf.

8. Plitt, Amy (2017, June 28). NYC's subway derailment in Harlem was caused by 'human error': MTA. Curbed New York. Retrieved from https://ny.curbed.com/2017/6/28/15885650/new-york-subway-harlem-derailment-cause.

SOURCES (cont.)

9. Winsor, Morgan (2017, June 28). Human error caused New York City subway train derailment, transit officials say. ABC News. Retrieved from https://abcnews.go.com/US/human-error-caused-york-city-subway-train-derailment/story?id=48322625.

10. Santora, Marc, and Fitzsimmons, Emma G. (2017, June 27). Subway Derailment in Manhattan Injures Dozens. *New York Times*. Retrieved from https://www.nytimes.com/2017/06/27/nyregion/subway-train-derails-in-manhattan.html.Rivoli, Dan (2018, July 6). Harlem subway derailment cost MTA $3.4M in new train cars, track work: report. *New York Daily News*. Retrieved from https://www.nydailynews.com/new-york/ny-metro-a-train-derailment-harlem-damages-20180713-story.html.

11. Honan, Katie (2017, June 28). TIMELINE: A History of Recent NYC Train Derailments. DNAinfo. Retrieved from https://www.dnainfo.com/new-york/20170628/midtown/new-york-city-mta-lirr-nj-transit-derailments

12. Fitzsimmons, Emma G. (2017, June 28). Subway Derailment Inquiry Focuses on Track Repairs. *New York Times*. Retrieved from https://www.nytimes.com/2017/06/28/nyregion/subway-derailment-inquiry.html.

13. Barab, Jordan (2017, July 3). Blame the worker — The error of blaming incidents on "human error." *Industrial Safety & Hygiene News*. Retrieved from https://www.ishn.com/articles/106803-blame-the-worker-the-error-of-blaming-incidents-on-human-error.

14. Conklin, T. (2019). *5 Principles of Human Performance*. Santa Fe: PreAccident Media.

15. Tredgold, Gordon (2018, Jan. 22). Bad Leaders React, But Here's What Good Leaders Do. *Inc.* Retrieved from https://www.inc.com/gordon-tredgold/good-leaders-know-why-responding-is-better-than-reacting.html.

16. Conklin, T. (2018). *Pre-Accident Investigations: Better Questions—An Applied Approach to Operational Learning*. Boca Raton, FL: CRC Press.

GRAPHICS

All images marked "© Quanta Services, Inc." are courtesy of Quanta Services, Inc. or one of its affiliates.

APPENDIX

APPENDIX A—ONLINE LEARNING FOR THE CAPACITY MODEL

The Online Learning for The Capacity Model website provides valuable educational resources. The site is open to everyone. There is no registration requirement. Use it as a resource to refresh and extend your learning.

ACCESS THE WEBSITE

Go to: https://learning.thecapacitymodel.com.

ELEARNING COURSES

Courses are available in English, Spanish, and French. Options include:

- Understanding The Capacity Model
- STKY Controls for Crews
- Introduction to Learning Teams

TCM VIDEOS

The videos provide viewers with an introduction to several elements of The Capacity Model. Some options include Quanta Human Performance Principles, STKY, Quanta Energy Wheel, and Quanta's Safety Philosophy.

STKY RE-CREATION VIDEOS

The re-creation videos are based on actual STKY events. Use these videos to help reinforce concepts such as using the Energy Wheel, asking capacity questions, identifying preventive and mitigative controls, and classifying STKY events.

RESOURCES

The site's Resources section includes PDFs of printable informational brochures and job aids such as the STKY Discussion Guide.

QUANTA

NLC